He does so in order to demonstrate that their number is large but finite, and can be determined. The numerical system of antiquity did not allow for dealing with very large numbers easily. In *The Sand Reckoner*, Archimedes develops a new system of numbering, similar to our exponentials, that makes it possible to deal with very large numbers, and shows its power by counting (certainly playfully) how many grains of sand there are, not just on the seashores but in the entire universe.

The Sand Reckoner is playful, but profound. With a flight of fancy that seems to anticipate the Enlightenment by millennia, Archimedes rebels against the form of knowledge that insists on there being mysteries which are *intrinsically* inaccessible to human thought. He does not claim to know the exact dimensions of the universe, or the precise number of grains of sand. It isn't the completeness of his knowledge that he is asserting. On the contrary, he is explicit about the approximate and provisional nature of his estimates. He speaks about possible alternatives regarding the true size of the universe – between which he does not make a definite choice. The point at stake here is not the presumption of knowing everything. It is the opposite: an awareness that yesterday's ignorance may have light shed on it today, and that today's might be illuminated tomorrow.

The central point is rebellion against the renunciation of the desire to know: a declaration of faith in the comprehensibility of the world, a proud retaliation to those who remain satisfied with their own ignorance, who call infinite that which we don't understand and delegate knowledge to elsewhere.

Centuries have passed, and the text of *Ecclesiasticus*, along with the rest of the Bible can be found in countless homes, while Archimedes' text is read only by the few. Archimedes was slaughtered by the Romans during the sacking of Syracuse, the last proud remnant of Magna Grecia to fall under the Roman yoke, during the expansion of that future empire which would soon adopt *Ecclesiasticus* as one of the foundational texts of its official religion, a position which it was to occupy there for more than a thousand years. During that millennium, the calculations made by Archimedes languished in a state of incomprehensibility: no one was able to use, or even to understand them.

Near Archimedes' Syracuse there is one of the most beautiful sites in Italy, the theatre of Taormina, which looks out at the Mediterranean and upon Mount Etna, the smoking volcano. In Archimedes' time, the theatre was used to stage plays by Sophocles and Euripides. The Romans adapted it for gladiatorial combat, for the pleasure of watching gladiators die.

The sophisticated playfulness of *The Sand Reckoner* is perhaps not only about an audacious mathematical construction, or the virtuosity of one of the most extraordinary minds of antiquity. It is also a defiant cry of reason, which recognizes its own ignorance but refuses to delegate to others the source of